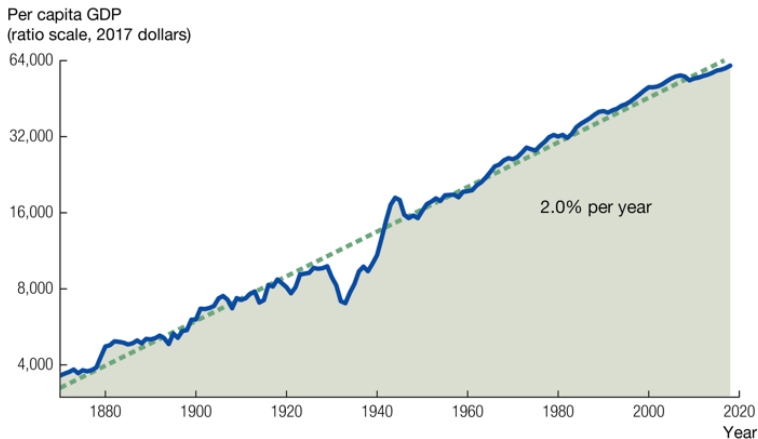


Macroeconomic Theory and Policy: Lecture 4 (Ch.3 and Ch. 4)

The Ratio (Logarithmic) Scale

- From time to time, you may see a plot where the tick marks are not of equal size



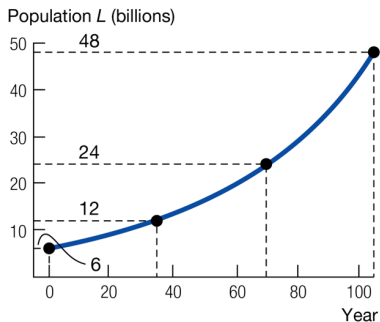
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The Ratio (Logarithmic) Scale

- A plot intervals between tick marks are proportional in terms of ratios rather than absolute differences
 - ▶ When doubling, we want this constant ratio to be "2."
 - ▶ e.g., on the y-axis, instead of "1, 2, 3, 4" we have "1, 2, 4, 8"
- When plotted on a ratio scale, a variable that grows at a constant rate will be a straight line
- It is commonly used to represent data where exponential growth or significant variations need to be visually compressed
 - ▶ Using the population growth figure earlier, we cannot tell right away that the growth rate is constant, increasing, or decreasing

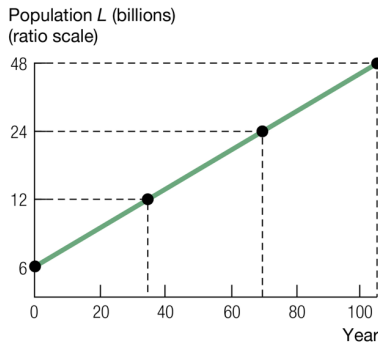
Population over Time, Revisited

- Absolute population numbers are increasing faster (left), but the rate of growth is staying the same in the right graph because of the linear relationship (right)



(a) On a standard scale...

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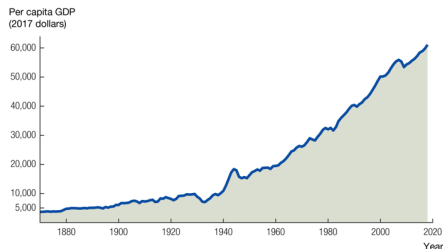
(b) and a ratio scale.

U.S. GDP on a Ratio Scale

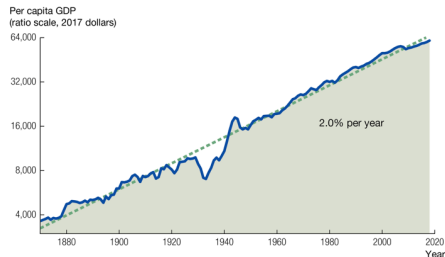
- On a ratio scale:
 - ▶ if growth rates are rising, the slope will be increasing
 - ▶ if the growth rate is constant, it will be a straight line
 - ▶ if the growth rate is falling, the slope will be decreasing
- Per capita GDP in the United States has grown at approximately 2 percent per year over the past 150 years.
 - ▶ This outcome is easy to see using a ratio scale
 - ▶ Approximately linear

Per Capita GDP in the United States, Standard vs. Ratio Scale

- The graph on the left is a standard scale and the one on the right is a ratio scale
- Data series lies close to a line that exhibits a constant slope of 2.0 percent per year
- To a first approximation, per capita GDP in the United States has been growing at a relatively constant annual rate of 2 percent over the past 150 years



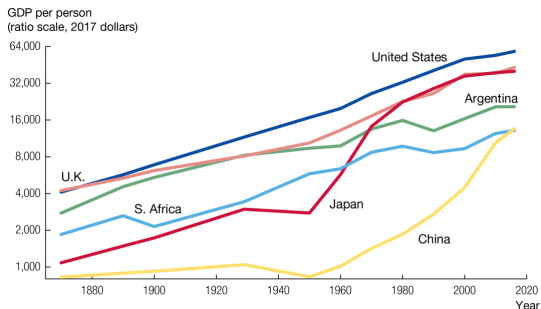
Source: Data from 1870 to 1928 from Barro- Ursua Macroeconomic Data, 2010. Data from 1929 to 2018 from U.S. Department of Commerce, Bureau of Economic Analysis.
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The Ratio Scale in Action: Per Capita GDP since 1870

- The picture contains more information, but we have to read it carefully
- The per capita income line is flat for the UK relative to the US until the 50s
- From 1880 to 1920, Argentina grew rapidly; Since 1980, growth has slowed considerably



Source: The Maddison-Project, www.ggd.net/maddison/. Observations are presented every decade after 1950 and less frequently before that as a way of smoothing the series.
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Modern Growth around the World

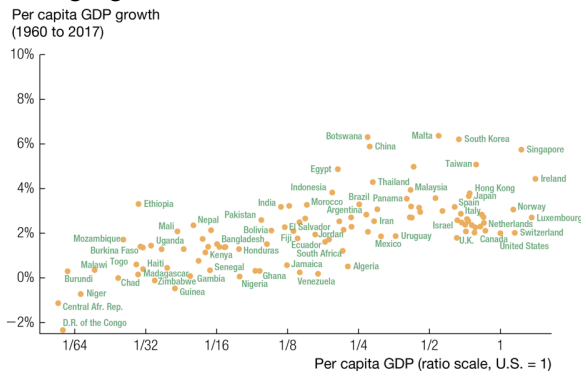
- In the late nineteenth century, the United Kingdom was the richest country in the world, but it started growing substantially more slowly than the United States in the 20th century
- After World War II, growth in Germany and Japan were similar, in that they both accelerated
- Argentina had accelerated growth until 1980 and then slowed considerably
- China, Vietnam and India have started “converging” recently
- **Convergence:**
 - ▶ Poorer countries will grow faster to “catch up” to the level of income in richer countries

A Broad Sample of Countries

- Over the period 1960–2017:
 - ▶ Some countries have displayed negative growth rates.
 - ▶ Other countries have maintained growth rates of nearly 7%.
 - ▶ Most countries have maintained approximately 2% growth rates.
 - ▶ Small variances in growth rates lead to significant differences in standards of living.

Levels and Growth Rates of Per Capita GDP

- From 1960 to 2017, growth rates ranged from -2 percent to +7 percent per year
- South Korea, Singapore, Botswana, and Japan grew the fastest over this period
- Central African Republic, the Democratic Republic of the Congo, Niger, and Guinea all had negative average growth

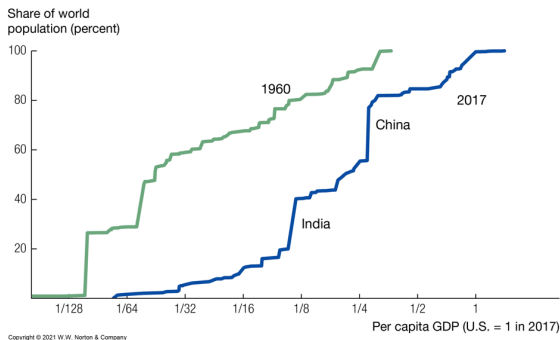


Source: Penn World Tables, Version 9.1. See the "Country Snapshots" file, snapshots.pdf, available from the author's web page for these data. The level of per capita GDP is taken from the year 2017 and is normalized so that the U.S. value equals 1.

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Case Study: People versus Countries

- Compare 2017 to 1960, the bulk of the world's population is substantially richer and the fraction of people living in poverty has fallen
- A major reason for changes:
 - ▶ Economic growth in China and India
 - ▶ These two countries account for 40 percent of the world population



Some Useful Properties of Growth Rates

- A lot of time we have economic variables express in term of some mathematical combination of other variables (e.g., GDP per Capita)
- Growth rates of ratios, products, and powers follow several (approx.) simple rules
- Suppose two variables x , y and z have average annual growth rates of g_x , g_y and g_z , respectively. Then

$$\text{If } z = x/y, \text{ then } g_z = g_x - g_y$$

$$\text{If } z = x \times y, \text{ then } g_z = g_x + g_y$$

$$\text{If } z = x^a, \text{ then } g_z = a \times g_x$$

- We can combine these rules

Examples of Growth Rate Calculations

TABLE 3.1

Examples of Growth Rate Calculations

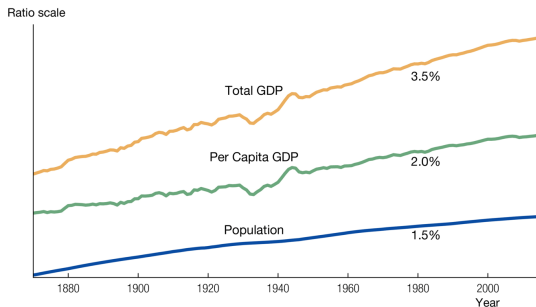
Suppose x grows at rate $g_x = 0.10$ and y grows at rate $g_y = 0.03$.
What is the growth rate of z in the following cases?

$z = x \times y$	$\Rightarrow$	$g_z = g_x + g_y =$	
$z = x/y$	$\Rightarrow$	$g_z = g_x - g_y =$	
$z = y/x$	$\Rightarrow$	$g_z = g_y - g_x =$	
$z = x^2$	$\Rightarrow$	$g_z = 2 \times g_x =$	
$z = y^{1/2}$	$\Rightarrow$	$g_z = 0.5 \times g_y =$	
$z = x^{1/2}y^{-1/4}$	$\Rightarrow$	$g_z = 0.5 \times g_x - 0.25 \times g_y =$	

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U.S. Population, GDP, and Per Capita GDP

- Total GDP = GDP per capita $\times$ number of workers,
- Growth rate of total GDP = growth rate of per capita GDP + the growth rate of the population



Sources: Angus Maddison, "Statistics on World Population, GDP and Per Capita GDP, 1 AD–2006 AD," and the Bureau of Economic Analysis.
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The Benefits and Costs of Economic Growth

- The benefits of economic growth:
 - ▶ Improvements in health
 - ▶ Higher incomes
 - ▶ Increase in the variety of goods and services
- Costs of economic growth include:
 - ▶ Environmental problems
 - ▶ Income inequality across and within countries
 - ▶ Loss of certain types of jobs
- Economists generally have a consensus that the benefits of economic growth outweigh the costs.

What are You Going to Learn (ch. 4)

- How to set up and solve a macroeconomic model
- The purpose of a production function and its use for understanding differences in GDP per capita across countries
- The role of capital per person and technology in explaining differences in economic growth
- The relevance of “returns to scale” and “diminishing marginal products.”
- How to look at economic data through the lens of a macroeconomic model

Introduction

- Recalling that a model:
 - ▶ is a mathematical representation of a hypothetical world that we use to study economic phenomena
 - ▶ consists of equations and unknowns with real-world interpretations
 - ★ Mathematically, a model is no more than a set of equations
- Economists ...
 - ▶ Document facts
 - ▶ Build a model to understand the facts
 - ▶ Examine the model to see how effective it is

A Model of Production

- Even though Model is only a simplifications of the real world, a good model can still provide important insights
- Consider the following model:
 - ▶ Single, closed economy
 - ▶ One consumption good
- Inputs in the production process: Labor (L) and Capital (K)
- Production function: how much output (Y) can be produced given any number of inputs

Setting Up the Model – Cobb-Douglas Production Function

$$Y = F(K, L) = \bar{A}K^{\alpha}L^{\beta}$$

- The F notation states that *output* (Y) is a function of *inputs* (K and L) - a general functional form
- $\bar{A}$ represents the productivity parameter
 - ▶ Letters with a bar over them denote parameters that are fixed and exogenously given in the model. These variables stand in for some fixed constant
- A higher value of $\bar{A}$ means the firm produces more of the output good with the same amount of inputs
 - ▶ The bigger the value of the productivity parameter, the more productive the firm.

Example: Growth Rules with Cobb Douglas Production Function

- Applying rules of growth rates to a Cobb Douglas production function:

$$Y_t = A_t K_t^{1/3} L_t^{2/3}$$

- Its growth rate of output can be decomposed into growth rate of a productivity term A , contributions to growth from capital K , and labor L
- Using If $z = x \times y$, then $g_z = g_x + g_y$,

$$g(y) = g(A) + g(K^{1/3}) + g(L^{2/3})$$

- Then, use If $z = x^a$, then $g_z = a \times g_x$,

$$g(y) = g(A) + \frac{1}{3}g(K) + \frac{2}{3}g(L)$$

Model

$$Y = F(K, L) = \bar{A}K^{\alpha}L^{\beta}$$

- Output growth corresponds to changes in Y
- There are three ways that Y can change:
 - ▶ Capital stock (K) changes
 - ▶ Labor force (L) changes
 - ▶ Ability to produce goods with given resources (K, L) changes
 - ★ Technological advances occur (changes in A)
 - ★ is assumed to be exogenous in the Solow model (a type of growth model)

Total Factor Productivity (TFP)

- Total factor productivity (TFP or Solow residual)
- Refers to all inputs that affect output (Y) except labor and capital
- This is the unexplained part of the production function, for example, human capital, energy, land
 - ▶ It is unexplained in the sense that it is not part of the function or be able to observed directly in the data
- An important assumption of the Solow model is that TFP is exogenous
- The model makes no effort to explain why TFP equals a particular value
 - ▶ This is reasonable if we believe that neither policy nor the choices of agents in the model affect its value
 - ▶ However, later we will discuss reasons this is likely a faulty assumption

More on Cobb-Douglas Production Function

$$Y = F(K, L) = \bar{A}K^{\alpha}L^{\beta}$$

- The parameter α and β determine the output elasticities of capital and labor
- In an usual setup, it also determine the share of labour input and capital output
 - ▶ The α is usually assume to be $1/3$
 - ▶ Recall that the labour income share is roughly $2/3$
- $F(K, L)$ is increasing in K and L
 - ▶ More inputs yield more output
 - ▶ Formally, $\frac{\partial F}{\partial K} > 0$ and $\frac{\partial F}{\partial L} > 0$

Constant Returns to Scale (CRS)

- A production function exhibits constant returns to scale if doubling each input exactly doubles output
- In the other word, if both K and L increase by $a\%$, then Y also increase by $a\%$
- Mathematically,

$$F(aK, aL) = aF(K, L)$$

i.e., Homogeneous function (of degree 1)

- Usually, we want the production function to be CRS. To do that we restrict the parameter to

$$\beta = 1 - \alpha \text{ and } 0 < \alpha < 1$$

- The production function became

$$Y = AK^\alpha L^{1-\alpha}$$

Constant Returns to Scale (CRS)

- There are other form of return to scale
 - ▶ If $\alpha + \beta > 1$, the function is increasing return to scale
 - ▶ If $\alpha + \beta = 1$, the function is constant return to scale
 - ▶ If $\alpha + \beta < 1$, the function is decreasing return to scale
- Constant returns to scale guarantee that as you increase the units of labor or capital required to produce your products, you maintain a consistent level of growth (more on that in later chapters)

Output per Person (Intensive Form)

- Divide output by the number of workers

$$\frac{Y}{L} = F\left(\frac{K}{L}, \frac{L}{L}\right) = F\left(\frac{K}{L}, 1\right)$$

- In this model, we assume the number of person = number of worker
- Lowercase letters denote per capita

$$y = f(k)$$

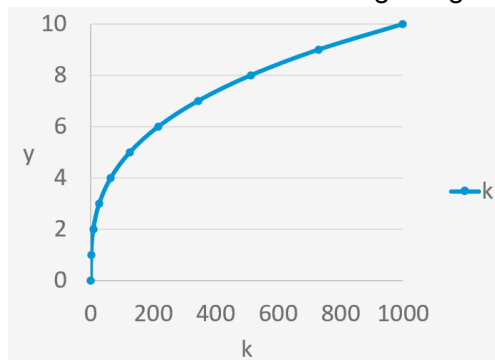
where

$$y = Y/L$$

$$k = K/L$$

Output per Person (Intensive Form)

- Plot of $y = f(k) = k^{1/3}$, where $y = 0$ when $k = 0$
- Growth models tend to explain deviations in output using changes in K or k
- Increasing and features diminishing marginal returns to capital
- Production function that is CRS can have diminishing marginal returns in K (or k)



Next Lecture

- Next lecture will finish covering Ch.4